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Supporting FLNG Design, Commissioning, and Start-Up with Dynamic Simulation Studies

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Abstract

Floating Liquefied Natural Gas (FLNG) facilities are increasingly recognized as a flexible, cost-effective means to develop remote gas reserves and deliver LNG directly to global markets. This paper presents the application of an advanced dynamic simulation study to support the design validation, commissioning, and start-up of a new 2.4 Mtpa FLNG facility. The study encompassed full production system modelling, with emphasis on liquefaction, cooling, fuel gas, and cargo handling systems.

By simulating transient scenarios such as start-up, shutdown, emergency shutdown, and trips of critical equipment, the study delivered actionable insights to optimize operational procedures, validate safety systems, and provide design bases for critical setpoints. The work enhanced operational readiness, reduced risks, and strengthened confidence in the facility's ability to transition safely and efficiently into commercial service.

Introduction

The development of offshore liquefied natural gas production technology has opened new opportunities for monetizing stranded gas reserves. FLNG facilities integrate gas treatment, liquefaction, storage, and offloading functions on a single floating unit, eliminating the need for long subsea pipelines and extensive onshore infrastructure. This reduces capital investment and environmental footprint while enabling flexible deployment in remote or challenging environments.

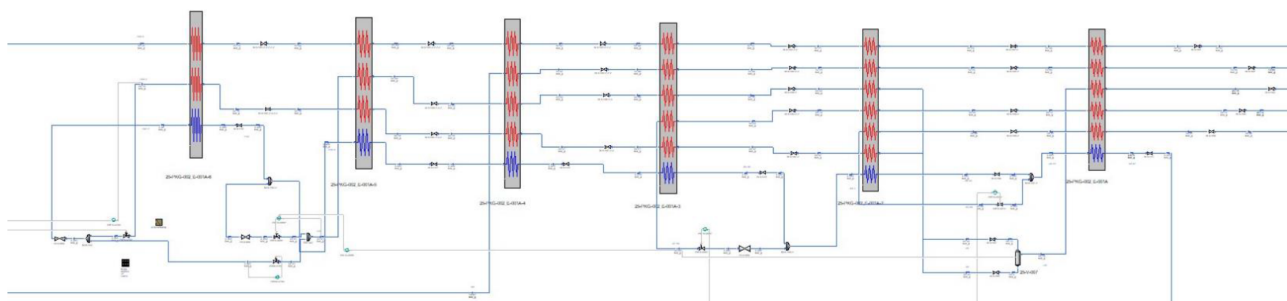


Figure 1—extract from part of the FLNG model showing the cold box and part of the liquefaction loop of one of the units

Technical challenges are significant: FLNG facilities must operate in dynamic offshore environments, handle variable feed gas compositions, and maintain reliability under diverse transient conditions. The integration of multiple process units — each with unique control systems, safety interlocks, and operational limits—requires thorough validation to ensure safe, stable, and efficient operation.

Dynamic process simulation has become an essential tool for such validation. Unlike steady-state models, dynamic simulation replicates time-dependent behaviors, allowing engineers to test start-up and shutdown procedures, control strategies, and emergency responses before operations. This improves safety, reduces commissioning time, and limits costly offshore interventions.

This FLNG unit will be stationed using a submerged swivel yolk mooring system within a marine field area, positioned approximately 50 km offshore in water depths around 30m. It is designed to handle around 380 MMscfd of gas from proximate fields, processing this input to produce, store, and offload LNG for exportation. This paper describes a dynamic simulation study conducted collaboratively between the FLNG contractor and simulation specialists to reduce operational risks and optimize readiness for commercial operation, ultimately supporting schedule certainty and long-term project value realization.

Methodology

Scope of Simulation

The dynamic simulation model encompassed the complete topside process systems, including:

- **Liquefaction trains** (main cryogenic heat exchangers, refrigerant compressors, and associated control systems)
- **Cooling systems** (seawater cooling including 8 pumps, closed-loop cooling including 6 pumps, as well as auxiliary systems)
- **Fuel gas supply system**
- **Cargo handling system** (LNG storage tanks, pumps, and offloading arms)

The dynamic simulation will encompass an integrated system simulation from the feed gas inlet facility unit through to the Cargo Tank and Boil-Off Gas (BOG) system. The main units included in the study are detailed as follows: Inlet facilities, Amine treating, Dehydration and Mercury Removal, NGL extraction and Feed Gas Booster, Liquefaction (Train I and II), LNG End flash drum, Refrigerant Make-up, BOG Compression, Condensate stabilization, Main Power Generation, Hot Oil system, Fuel Gas system, Fresh Cooling Water distribution system, LNG Cargo and Handling system 1-2-3-4, LPG Cargo and handling system, Condensate Cargo and handling system, Sea Water system, Fresh water cooling system, Regas commissioning skid.

This includes an array of process and utility units — specifically, the cooling water system, fuel gas system, and hot oil system — during initial start-up, and across a variety of transient scenarios. The study

aims to verify the system's capacity to withstand transient conditions and achieve a new stationary state during system initial start-up, operating upsets, and restart scenarios.

The model also incorporated utilities and safety systems relevant to transient operations, ensuring that critical interactions between systems could be evaluated. The results obtained with the dynamic simulation model, once stable in steady state, have been compared against the H&MB provided by Wison. The relative or absolute error between the main selected variables (i.e. Molecular Weight, Mass flow, Temperature, Pressure, Density...) is then calculated.

Furthermore, an additional validation against manufacturers Data Sheet information is carried out as well for all critical equipment. This ensures consistency with design H&MB and avoids gaps on design or equipment operational limitations for the envisaged scenarios. Model accuracy shown to be greater than 95% for almost all variables and reducing the discrepancies with design data to below 2% for the most critical variables. Furthermore, combination of different fluid packages with the required component lists and EOS, allowed to properly tackle each operating unit needs to keep overall model accuracy. Once the commissioning phase occurs after vessel sails away from yard, a support to commissioning and start-up support activity is envisaged (not yet done).

Simulation Platform

The study was implemented using Aspen HYSYS® Dynamics, a widely adopted process simulation platform capable of representing equipment-level details, thermodynamics, hydraulics, and control system behavior. The model included, among others:

- Detailed compressor and pump performance maps
- Anti-surge and complex control logic
- Valve and actuator dynamics
- Heat exchanger thermal exchange, including complex systems such as cold box
- Tank accumulation and handling of boil-off gas generation

Process Flow Diagram (PFD) sub flowsheet capabilities were used to build models that could handle different compositions and fluid packages as required for each operating unit. This allowed Inprocess team to enhance model capabilities according to the purpose of the scenarios to be run in each one. Hence, bigger integrated models were optimized to address the holistic approach of detecting the interactions and general implications between the different operating units while smaller single models of each unit were used for detailed analysis of inherently faster scenarios such as surge excursion detection where a lower time step is needed and greater calculation demand from the model is required. The dynamic model is based on first principles mathematical modelling thus, the equipment parameters, the flow resistances, the heat transfer coefficients and the fluid accumulation are determined based on the equipment geometry and size.

This allows representing and adjusting the system responses in accordance with equipment dimensions, in order to accurately represent the plant hydrodynamics, with the necessary level of detail to properly carry out the training scenarios. Isometric analysis and implementation was as well carried out in order to include into the model the details for accurate volume hold-up calculation in each section including the actual expected ΔP due to the pressure loss in the different fittings and due to the length of the piping.

Scenarios Studied

A series of operational scenarios were defined jointly by the contractor's engineering team and the simulation specialists, based on design documentation, commissioning plans, and safety studies. These included:

1. **Initial start-up.**
2. **Restart** of critical equipment after unexpected shutdown.
3. **Planned shutdown** under normal operation to ensure smooth transient up to no-flow condition.
4. **Emergency shutdown (ESD)** initiated by safety system activation.
5. **Trips of critical equipment**, such as refrigerant compressors, Compander and booster compressor, to test process resilience and equipment integrity based on process conditions.
6. **Transient load changes**, including ramp-up to maximum capacity and ramp-down to minimum turndown as well as sudden feed gas losses.

For each scenario, operational sequences were validated, control system responses were assessed, and process/equipment integrity was verified. Scenarios were not simply analyzed, but a thorough revision of the possible mitigation actions that could lead to the plant survivability was conducted hand to hand with Wison Engineering and operations team. The first goal was to identify the required actions, and then, through an iterative approach, those actions were combined in time and reduced to the minimum possible to ease actual operator actions ensuring the risk and failure reduction to the minimum possible due to lack of response time, failure to implement certain action, confusion in the order of the actions to be implemented or skipping any of the required actions. With this basis, operation procedures were updated, and base cases were showcased to anticipate actual operating scenarios to which the operators shall be trained on.

For the scenario analysis, it was proceeded with 2 approaches: on the one hand, Inprocess used a general integrated model to be able to capture interactions between the main units, as the actions and mitigations posed in one unit may affect upstream/downstream units, leading to side effects to be handled as well. This provided a general overview, very useful to keep all the results aligned and consistent within the different operating units. On the other hand, specific scenarios such as ESD or sudden blockage/closure of main valves were performed for secoific units/equipment. This provided very detailed insights on equipment performance (sudden pressure peaks, temperature changes, flow distribution...) to ensure that the integrity was kept, impacting positively in overall maintenance and equipment aging over time.

Results

While specific numerical results are confidential, the study delivered clear operational and safety benefits:

- **Validation of operational procedures:** Start-up, restart, and shutdown sequences were tested in detail, confirming that the control logic, valve sequencing, and interlocks performed as intended. Suggestions on improved steps and assessment on specific set points or manual valve openings were provided as well. Some of the main topics discussed were the avoidance of stonewall excursion during start-up, for which recommendations like further closing of ASV at specific points of the procedure were suggested, including shifting of acceleration of compressor and halt in opening of other process control valves were applied, to ensure smooth operation throughout the whole procedure within the operable limits.

For ESD conditions, avoidance of surge and big temperature or pressure peaks were to be avoided. To do so, some cases require dampening certain effects by preemptively opening blowdown valves or acting in some controllers in advance. Also, fine tuning of ASC was provided to ensure smooth and safe operation. Of course, this tuning along with the rest of control parameters for all the controllers is to be confirmed later on during commissioning and start-up activities based on actual compressor controller algorithm and parameters, but simulation results give a solid basis to understand the capacitance of the system and how it handles different upsets.

- **Verification of safety system performance:** Overall ESD activation was shown to bring the plant to a safe state without exceeding design limits. A good operational asset is not only the one that operates steady and close to an optimal production, but the one that handles correctly the upsets

and the unplanned events such as an overall trip. In this case, the whole plant was subjected to an Emergency shutdown followed by a EDP Blowdown. The objective was to check that each unit could be driven towards no flow condition without major peaks on the main variables.

In the figure below, the LNG product temperature can be observed, serving as an example of a steady and bump-less performance, which was one of the main objectives. It is worth noting that for this scenario almost no additional reruns were needed, nor mitigation actions. Another figure is provided, showing a smooth coast down of a compressor during a controlled shutdown. In this way, the sequence of events defined by Wison was tested and confirmed that actual procedures were working as expected. This indicates the high level of quality provided by Wison Engineering team, in which the equipment and overall process responded very well to a major trip of the whole plant.

- **Equipment robustness assessment:** The system was shown to maintain integrity when subjected to sudden trips of major equipment, avoiding surge, excessive pressure drops, or uncontrolled temperature swings. Figure below shows specific trends for one of the main cold box layers. One of Wison's concerns about operation of the asset was not only the survivability of the process itself, but how the operating conditions may affect the robustness and integrity of the equipment. In this case, the temperature changing trends in the event of an ESD for a cold box were tracked. The main objective was to ensure that the change of temperature in time did not exceed manufacturer recommended limits to avoid thermal stress to the material of the layers.
- **Plant survivability:** dynamic analysis allowed studying the mitigation actions needed to dampen the effect of critical equipment trip or failure, allowing the FLNG to survive in the event of different upset conditions. In the below figure a major upset of 50% Feed Gas Loss to the plant is studied. One of the first alarms to be triggered is a PALL located at the entry pipe. Inprocess needed first to observe the whole transient in the plant. This allowed to identify the main controllers playing a key role in making the plant survive and then start testing how many mitigation actions shall be taken and at which time, so to ensure that the whole plant could be driven towards a new steady state condition at a lower production.

Since the upset here happens almost immediately, fast and coordinated actions are to be taken. In this case, the collaboration with Wison engineering and operations team was paramount in order to identify the feasibility of the actions that could be taken or not, as well as the effectiveness of those for the plant correct operation (integrity of rest of units, product quality, amount of flaring, etc.). The nature of this scenario is quite iterative, as certain mitigation actions may lead to different collateral effects and other outcomes, but at the same time, provide a clear overview on how the plant and different units are performing, allowing to find ways for improvement.

- **Design basis confirmation:** Certain critical setpoints, such as compressor recycle valve openings and cooling water temperature limits, were validated through simulation, providing a robust basis for operational documentation. Additional actions such as flaring and flow decrease in the event of certain equipment trip, understanding the timing and the combined actions from operational perspective were key to strengthening the operational procedures for a better and safer action.
- **Safety and Lag time responses:** dynamic analysis allowed to confirm the suitability of the safety actions and elements designated for critical trips, ensuring that the response came in time and within the operating range needed. Main equipment studied: Compander ASV opening failure, Stabilizer reflux pump trip, Cooling water loss, regeneration gas cooler failure, blocked inlet of one liquefaction train, loss of Natural Gas from NGL extraction unit, Amine pumps failure, etc.

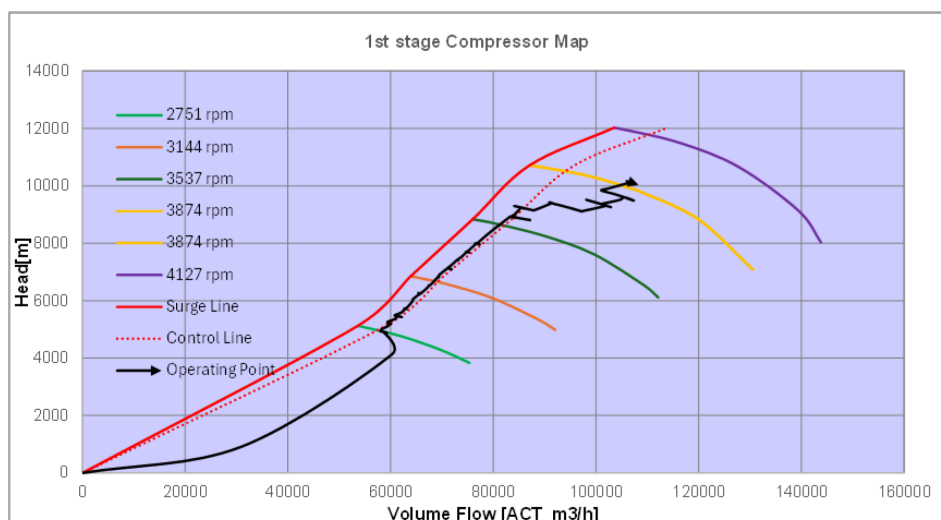


Figure 2—Start-up check of MRC

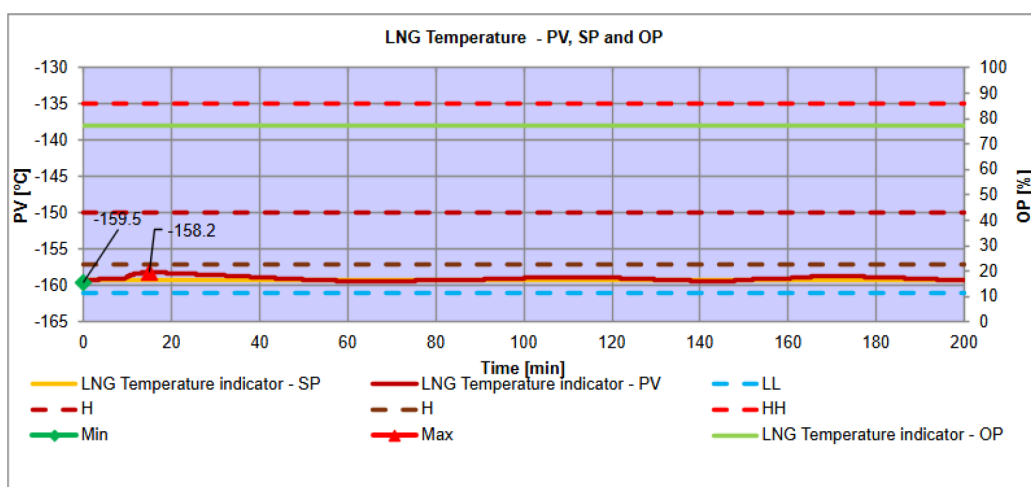


Figure 3—critical variable control during plant operational changes to ensure operation within safe limits

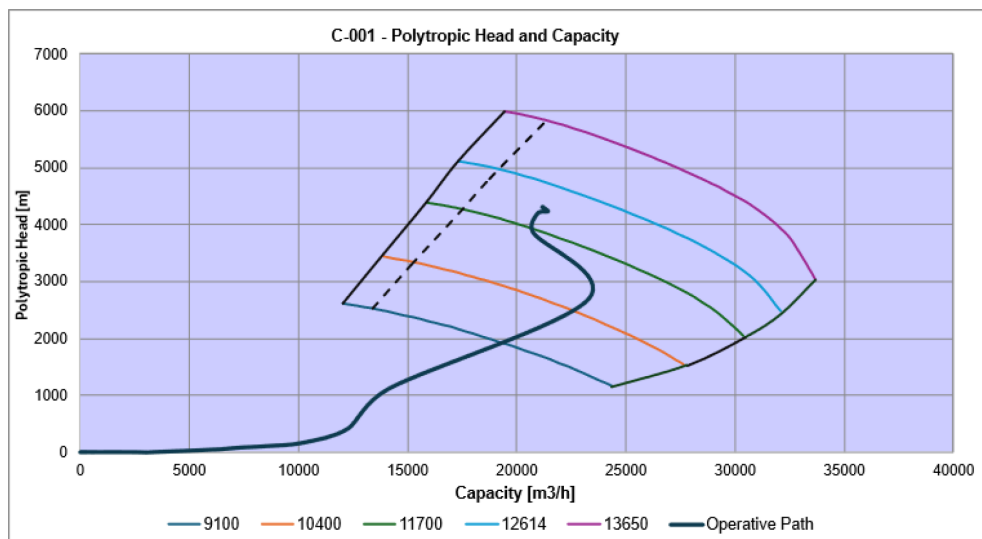


Figure 4—compressor coast down without surge excursion during a controlled shutdown

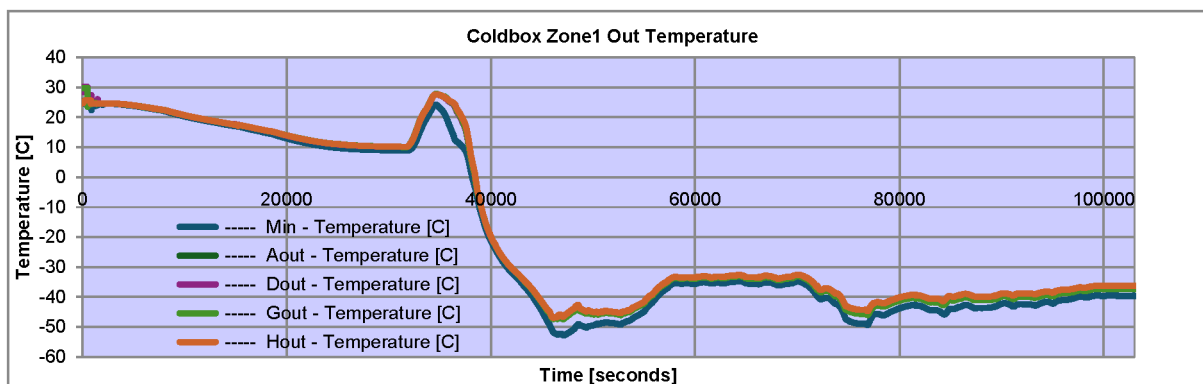


Figure 5—coldbox temperature zone change analysis for assessment on equipment integrity

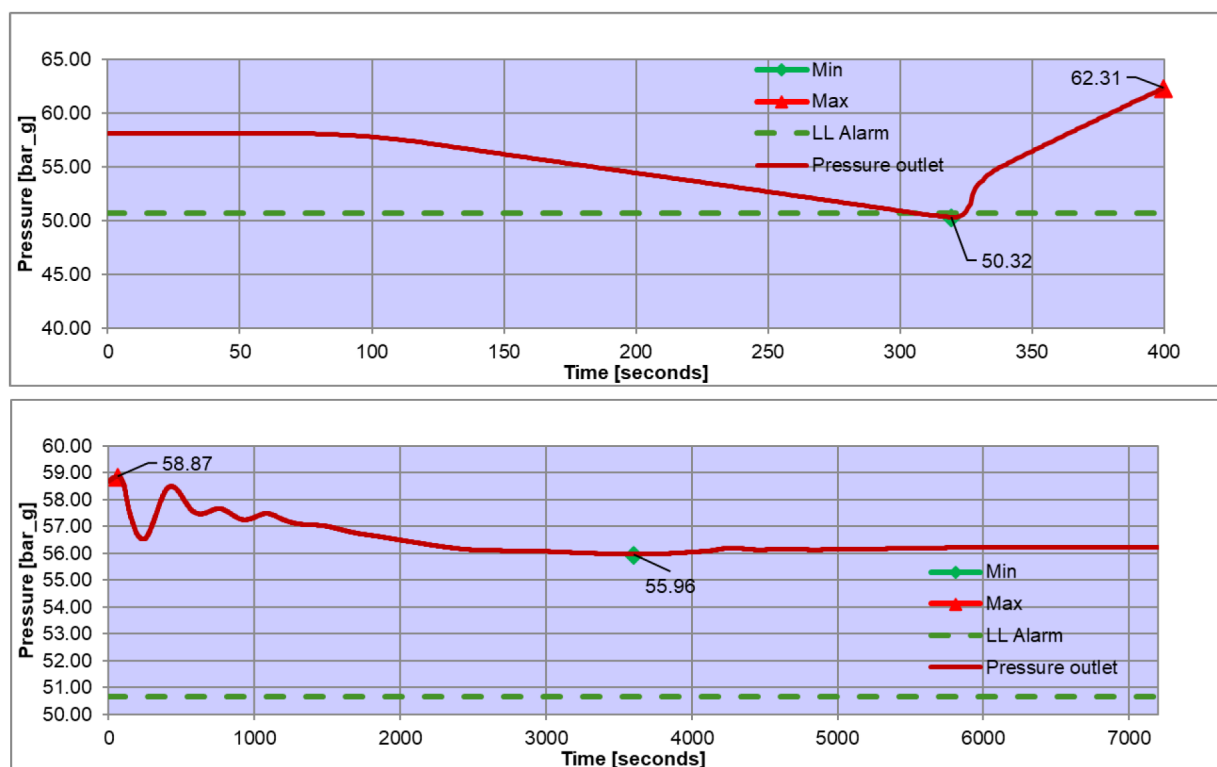


Figure 6—Plant survivability without/mitigation action for a low-low pressure trip

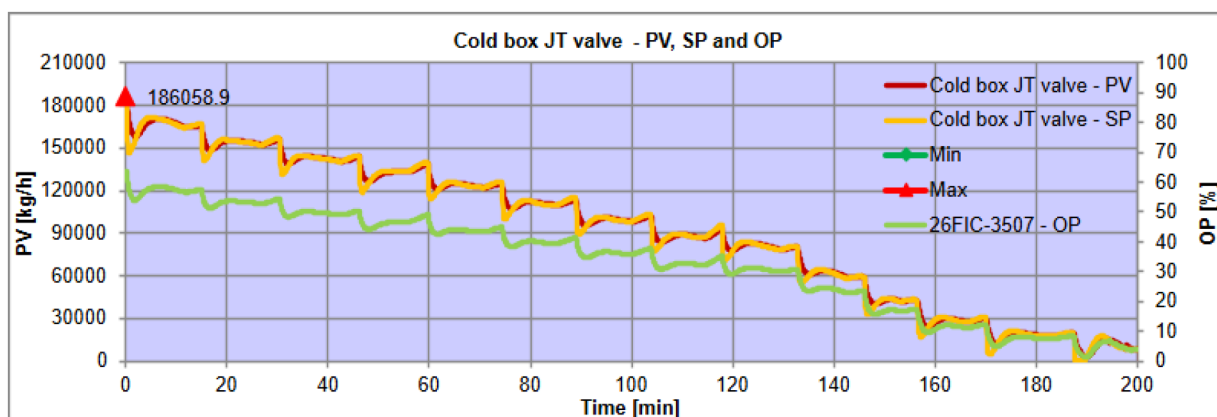


Figure 7—part of step-by-step procedure for turn-down condition and plant shutdown

These were provided to the EPC company to confirm the expected response both from field or from control/safety system to prevent equipment damage and ensure smooth transition towards safe operating conditions. Knowing these times allows the engineering and operations team to frame the reaction time window that they have and decide whether field, automated or controlled action is required in each unit to ensure that the plant can survive or overcome any potential upset.

Discussion

Dynamic simulation delivered multiple benefits:

- Reduced risk through early identification of bottlenecks, unstable control loops, and safety response delays
- Enhanced commissioning readiness by pre-validating sequences and setpoints
- Knowledge transfer by familiarizing engineers with system responses in a virtual environment
- Ensured equipment integrity under several upset scenarios in which not only was the process evaluated, but equipment design limits were challenged as well.
- Having the dynamic model as a test bed allows not only to check the scenarios in advance, but also to find ways to optimize and enhance control over the plant, knowing how disturbances are dampened by the system and how different control actions help to avoid them.
- Dynamic model analysis was used in parallel to OTS development by a third party. Outcomes from the study served as well to design the right training scenarios of the OTS and double check its performance, confirming a full understanding and alignment between engineering and operations team.

The approach aligns with industry moves toward digitalization and "virtual commissioning" for lifecycle support.

Conclusions

The dynamic simulation study was key to preparing the FLNG facility for safe, efficient operation. By replicating process dynamics and testing critical scenarios, the team:

- Validated operational procedures, reducing commissioning time and safety risks
- Confirmed equipment resilience, improving maintenance and extending life expectancy
- Established robust design bases for critical parameters, enabling efficient and profitable operation across varied conditions

Results of the study helped Wison engineering team to confirm their design basis, increase their confidence in the operating procedures, being able to improve them in advance and reducing the risk of unknown or unplanned situations during critical operations during the commissioning stage. This allowed Wison to keep the initial schedule without delays, ensuring that activities happen in accordance with plan and reducing overall the whole process from design to commissioning and operation.

The collaboration ensured both design intent and operational realities were addressed, creating a smoother path from project design to execution. By anticipating risks early and embedding flexibility into operational strategies, the study provided not only technical validation but also commercial value. As FLNG projects grow in complexity, dynamic simulation will remain central to de-risking operations, accelerating time to first LNG, and enhancing long-term plant performance.

Nomenclature

ASV	– Anti Surge Valve
DS	– Data Sheet
EDP	– Emergency Depressurization
EOS	– Equation Of State
ESD	– Emergency Shutdown
FLNG	– Floating Liquefied Natural Gas
H&MB	– Heat and Material Balance
LNG	– Liquefied Natural Gas
MMSCFD	– Million Standard Cubic Feet per Day
Mtpa	– Million Tonnes per Annum